

Comparative Analysis of Finnish and Hindi Corpora

Language as Data - Assignment 1

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1. Language Characteristics

Hindi

Name(s) of the language:

In English, the language is called “Hindi”. In the language itself the name is called “हिन्दी”. With the Hunterian transliteration system it is converted to the Latin alphabet as “Hindī” [1].

Number of speakers:

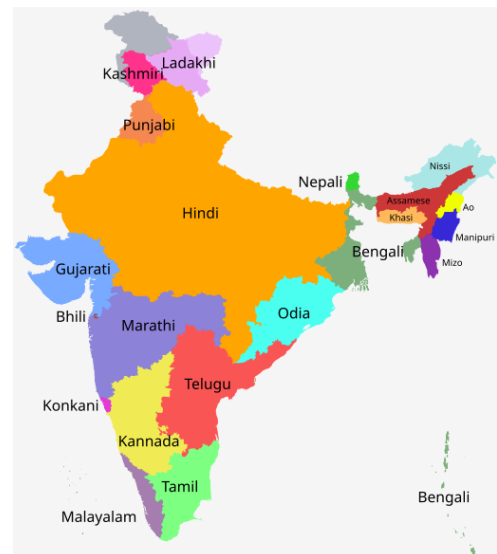
Around 600 million people speak Hindi. It has more than 350 million native (L1) speakers [2][3]. It is the language with 3rd most speakers and 2nd most L1 speakers [5].

Preservation status:

The language is not endangered. It is widely spoken as native language, and used for education and administration in India [4].

Geographic region:

Hindi is mainly spoken across northern and central India, with significant diaspora communities in the USA, Canada, Europe, as well as in Nepal, Mauritius, and Fiji [5]. Historically, Hindi is a direct descendant of an early form of Vedic Sanskrit, which originated in the northwestern Indian subcontinent [6].



Language family:

Hindi belongs to the Indo-European language family, within the Indo-Iranian → Indo-Aryan branch.

It can be counted as a part of Indian Sprachbund where languages such as Hindi and Punjabi from Indo-Aryan family, Tamil and Telugu from Dravidian family and Santali from Munda family have developed similar features due to contact [7].

Grammar:

The typical word order is SOV [8]. According to Feature 22A “Inflectional Synthesis of the Verb” of Wals, How “synthetic” a language is can be measured by counting categories per verb word (cpw).

English past tense walked = 2 cpw (tense + agreement)

Verbs in Hindi also have 2-3 grammatical categories such as tense, agreement per word.

According to Feature 26A “Prefixing vs. Suffixing in Inflectional Morphology”, Hindi is strongly suffixing, meaning that more than 80% of its total inflectional affixes are suffixes. This indicates that almost all grammatical markers (like tense, plural, case, or possession) are added at the end of words, rather than the beginning.

Hindi is aspect-split ergative, meaning the subject of a transitive verb is marked with the ergative case only in the perfective/past aspect, while in other aspects subjects are marked normally [10].

Hindi has a moderately high consonant-vowel ratio, it has 4.5 to 6.5 times as many consonants as vowel qualities [11]. The language has many consonants but relatively few distinct vowel sounds. It is not a tonal language, the tone does not change word meaning.

Hindi has three noun cases (nominative, oblique, and vocative) and five pronoun cases (nominative, accusative, dative, genitive, and oblique) [12].

Orthography

Hindi is written in the Devanagari script, an abugida. Devanagari consists of 11 vowels and 33 consonants and is written from left to right [13]. The Government of India uses Hunterian transliteration as its official system of writing Hindi in the Latin script [1].

Devanāgarī belongs to the Brahmic family of scripts, a group found across India, Nepal, Tibet, and Southeast Asia. It originated from the Brāhmī script of the 3rd century BCE, evolving first into the Nagari script, which subsequently led to the development of both Devanāgarī and Nandināgarī [14].

Finnish

Name of the Language

The native name of the language is “suomi”.

Number of Speakers

Approximately 5 million people use it as their native language. And around 500.000 people speak it as a second language in Finland [15].

Prevention Status

Finnish is not an endangered language.

Geographic Region

It is mainly spoken in Finland, and also some regions of Sweden and Norway.

Language Family

It is a Finnic language, a branch of the Uralic family. Some other Finnic languages are Estonian spoken mainly in Estonia, Karelian spoken in the state of Karelia in Russia, Veps spoken mainly in northwestern Russia, south of Karelia.

Also Meänkieli, a historical variety of Finnish spoken in Sweden and Kven spoken in Norway. These two are also intelligible to a Finnish speaker [17].

It is believed that the languages in the Circum-Baltic area have influenced each other through centuries. This area includes languages such as Finnish, Estonian, Baltic languages (Latvian, Lithuanian), Slavic languages (Russian), Germanic languages (Swedish) [18].

Grammar

The language follows an SVO order. According to features 22A & 26A of WALS, Finnish also has 2-3 categories per word [9] and it is strongly suffixing.

Finnish has three grammatical persons. First person – the speaker (I, we). Second person – the listener (you, singular or plural) Third person – someone/something else (he, she, they, it) [19].



Finnish is called an accusative language because it often marks the direct object with a special ending called the accusative case.

Example:

- Minä näen koiran → “I see the dog.”
- koiran (“dog”) has the -n ending, showing it is the object of the verb see.

Finnish only has two real tenses—present and past—but it can combine them with a special “perfect” construction. This gives four basic forms: present, past, present perfect, past perfect. Finnish doesn’t have a separate future tense. Instead, context and the way the object is marked tell whether something happens now or in the future [19]. This can also be seen in “Feature 67A: The Future Tense” from WALS, where Finnish is marked to have “no inflectional future [20].

Finnish has fewer “basic words” than English. Instead of having many separate words, Finnish builds new words by adding endings and suffixes to existing words [19].

According to “Feature 49A: Number of Cases” from WALS, Finnish has more than 10 cases, where a case is an ending added to a noun to show its role in the sentence — who does what to whom, where something happens, direction, etc. For comparison, English has only 2 cases [21]. Therefore, Finnish lets you express very precise relationships using compact forms:

- talo = house
- talossa = in the house
- talosta = from the house
- taloon = into the house
- talotta = without the house

Its consonant-vowel ratio is moderately low, meaning that the language has relatively few consonants compared to its vowel qualities. We can say that the language’s sound system relies more on vowel distinctions and less on consonant variety. It is also not a tonal language.

Orthography

Finnish uses the Latin script for its writing system, with an alphabet derived from Swedish. Officially, it consists of twenty-nine letters, with two extra letters appearing in certain loanwords [22].

a	b	c	d	e	f	g
h	i	j	k	l	m	n
o	p	q	r	s	t	u
v	x	y	z	å	ä	ö

2. Corpus Datasheet

Hindi Corpus Datasheet

Corpus Information

Corpus Name: Leipzig Corpora Collection - Hindi News 2022

Source: Wortschatz Leipzig [23][24]

Size: 100,000 sentences

Type: News corpus

Language Code: hin

Download: <https://wortschatz.uni-leipzig.de/en/download/hin>

Composition

Q1. What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)?

Each instance in the Leipzig Hindi News 2022 corpus represents a single sentence extracted from Hindi-language news websites. The sentences are self-contained linguistic units that were automatically segmented from longer news articles using sentence boundary detection algorithms [24].

Q2. How many instances are there in total (of each type, if appropriate)?

The corpus contains 100,000 sentences. The Leipzig Corpora Collection provides Hindi corpora in multiple standardized sizes (10,000, 30,000, 100,000, 300,000, and 1,000,000 sentences), allowing researchers to select corpus size based on their computational resources and research needs [23][24].

Q3. Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set?

This is a sample corpus, not an exhaustive collection. The data was collected from publicly available Hindi news websites via automated RSS feed monitoring during 2022 [24]. The sampling process involves random selection from available online Hindi news texts, which introduces several potential biases:

- **Genre bias:** Only news content is represented, excluding social media, blogs, literature, and conversational language
- **Source bias:** Limited to news outlets with RSS feeds and web presence, potentially excluding regional or local newspapers
- **Temporal bias:** Represents only 2022 language use, missing historical variation and language evolution

Q4. What data does each instance consist of?

Each instance consists of [23][24]:

- A unique sentence identifier
- The sentence text in Devanagari script
- Pre-computed word co-occurrence statistics for significant collocations
- Minimal metadata about source (typically limited)

The corpus does NOT include: original URLs, publication dates, author information, article titles, or document-level context.

Q5. Is any information missing from individual instances?

Yes, substantial information is missing:

- **Source attribution:** No original URLs, publication dates, or specific news outlet identification
- **Author information:** No journalist names or author metadata
- **Document context:** Sentences are extracted from articles without surrounding context
- **Genre/topic labels:** No classification of news type (politics, sports, entertainment, etc.)
- **Demographic information:** No data about writers or subjects mentioned
- **Quality indicators:** No reliability scores or fact-checking metadata

This missing information limits analyses of source diversity, temporal trends, authorial style, and topical distribution.

Q6. Are there any errors, sources of noise, or redundancies in the dataset?

Potential sources of noise and errors include [24]:

- **Language mixing:** Approximately 10% of collected web material may contain foreign language content despite language filtering based on the 5,000 most frequent words. Hindi news frequently includes English words and phrases (code-switching), which may or may not be filtered.
- **Sentence segmentation errors:** Automated sentence boundary detection may incorrectly split or merge sentences, especially with:
 - Abbreviations (e.g., "डॉ.", "श्री")
 - Unusual punctuation in headlines
 - Quoted speech within articles
 - Lists or enumerated items
- **Non-sentences:** Some extracted units may not be grammatically complete sentences:
 - Headlines and subheadlines
 - Photo captions
 - List items or bullet points
 - Navigational text from web pages

- **Duplicate removal:** While explicit duplicate sentences are removed, near-duplicates (e.g., slightly reworded agency reports published by multiple outlets) may remain.
- **Code-switching:** Hindi journalism commonly incorporates English words and phrases, particularly for technical, political, and international topics. The extent of filtering for such mixed-language content is not fully documented.

Q7. Does the dataset contain data that might be considered confidential?

Not applicable. All data was collected from publicly available news websites. The Leipzig Corpora Collection explicitly states that material is "taken from carefully selected public sources" [23]. However, news articles may quote or reference individuals, and while this information is public, researchers should be aware that:

- News content may include names of private individuals involved in news events
- Sensitive personal situations may be described (crime victims, accident victims, etc.)
- The corpus creators take no responsibility for the content, stating views remain with the original authors

Q8. Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?

Yes, potentially. News corpora inherently contain sensitive content [23]:

- **Violence and crime:** Reports of murders, assaults, accidents, disasters
- **Political controversy:** Descriptions of political conflicts, protests, riots
- **Social issues:** Coverage of discrimination, inequality, sensitive cultural topics
- **Negative events:** Natural disasters, epidemics, economic crises
- **Potentially biased language:** News sources may reflect particular political or cultural viewpoints

The corpus creators explicitly state: "No responsibility is taken for the content of the data. The views and opinions expressed in specific parts of the data remain exclusively with the authors" [23].

Q9. Does the dataset identify any subpopulations (e.g., by age, gender)?

Not findable in corpus metadata. The corpus does not contain systematic demographic annotations. While news articles may discuss various demographic groups, there is no structured labeling of:

- Which population groups are mentioned
- How frequently different groups appear
- The context in which groups are discussed
- Speaker or subject demographics

Q10. How was the data associated with each instance acquired?

The data was automatically collected through web crawling [24]:

- **News material:** Collected via RSS feeds from news websites on a daily basis
- **Collection method:** Automated using distributed web crawlers
- **Language identification:** Content filtered using a language identifier based on the 5,000 most frequent Hindi words
- **Random selection:** Sentences randomly selected from the collected pool to create standardized corpus sizes

Source selection bias: The method favors large, established news outlets with robust web infrastructure and RSS feeds, potentially underrepresenting smaller, regional, or independent media voices.

Collection Process

Q1. Over what timeframe was the data collected?

The "Hindi News 2022" corpus was collected during the year 2022. The Leipzig Corpora Collection creates annual releases, with news material collected via RSS feeds throughout the specified year. However, the documentation acknowledges that crawled material "may be older than the specified year" [23], meaning some content from late 2021 or archived articles may be included.

Q2. Were any ethical review processes conducted (e.g., by an institutional review board)?

Not findable. The Leipzig Corpora Collection documentation does not mention institutional review board (IRB) approval or formal ethical review processes. However, several ethical considerations are implicit in the methodology:

- Data is exclusively from public sources
- No personal data collection from individuals
- No first-person accounts or user-generated content requiring consent
- Follows general principle that publicly available web content may be collected for research

The lack of explicit ethical review documentation may reflect that the project predates current data ethics standards, or that European data collection practices differ from U.S. IRB requirements.

Q3. Does the dataset relate to people?

Yes, indirectly. News articles frequently discuss people:

- **Public figures:** Politicians, celebrities, business leaders, athletes
- **Private individuals:** People mentioned in news events (victims, witnesses, participants)
- **Collective groups:** References to communities, demographics, organizations

However, the corpus does not involve:

- Direct data collection from individuals
- First-person accounts or interviews requiring consent
- Personal data beyond what appears in published news

Privacy consideration: While information is from published news sources, the aggregation and permanent archival of names and events in a searchable corpus raises questions about the "right to be forgotten" and whether individuals mentioned in news have consented to their inclusion in research datasets.

Q4. Was there a specific task the dataset was created for?

The Leipzig Corpora Collection was created for general-purpose linguistic analysis and NLP applications, not for a specific downstream task [23][24]. Intended uses include:

- Word frequency analysis and lexicography
- Statistical language modeling
- Cross-linguistic typological comparison
- N-gram and collocation extraction
- Language resource development for under-resourced languages
- Corpus linguistics research

The standardized format and comparable sizes across languages specifically enable multilingual comparative studies.

Preprocessing/Cleaning/Labeling

Q1. Was any preprocessing/cleaning/labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)?

Yes, extensive preprocessing was performed [24]:

1. **Text extraction:** Raw HTML extracted from web pages and RSS feeds
2. **Sentence segmentation:** Text automatically split into sentences using language-specific sentence boundary detection algorithms that identify sentence-ending punctuation while handling abbreviations and special cases
3. **Language identification:** A language identifier based on the 5,000 most frequent Hindi words filters out foreign language content
4. **Non-sentence removal:** Pattern-based filtering identifies and removes strings that do not conform to expected sentence structure (e.g., very short fragments, navigation elements, repeated boilerplate)
5. **Duplicate removal:** Exact duplicate sentences are identified and removed from the corpus
6. **Random selection:** Sentences are randomly ordered and selected to create standardized corpus sizes
7. **Word tokenization:** Sentences are tokenized into words for frequency counting
8. **Co-occurrence computation:** Significant word co-occurrences (at sentence level) are pre-computed using log-likelihood ratio and included with the corpus

No annotations added: The corpus remains unannotated (no POS tags, syntax, or semantic labels)

Q2. Was the "raw" data saved in addition to the preprocessed/cleaned/labeled data (e.g., to support unanticipated future uses)?

No. The Leipzig Corpora Collection provides only the processed data (sentences after cleaning, segmentation, and selection). The original raw web pages, HTML content, or complete news articles are not distributed or archived as part of the corpus release [23].

Limitation: Researchers cannot:

- Verify sentence extraction accuracy
- Access full document context
- Recover article metadata
- Re-segment or re-process the raw data

Q3. Is the software used to preprocess/clean/label the instances available?

Partially available [24][25]:

- **Methodology documentation:** The general corpus production methodology is described in published academic papers, particularly Goldhahn et al. (2012)
- **Open-source tools:** Some NLP tools have been released on the Leipzig-Corpora-Collection GitHub repository, including sentence cleaners and language identification tools
- **TinyCC engine:** The corpus production engine (TinyCC) has been mentioned in documentation but full implementation details are not completely open-source
- **Database scripts:** MySQL import scripts are provided with corpus downloads

Reproducibility limitation: While the methodology is documented, the complete pipeline is not fully reproducible without access to the original web crawl data and proprietary components.

Q4. Any other comments?

Does this corpus achieve its intended purpose?

Yes, for its stated goals [23][24]:

- Provides standardized-size corpora suitable for cross-linguistic comparison
- Offers clean, sentence-segmented data usable for NLP applications
- Includes useful pre-computed word co-occurrence statistics
- Makes Hindi linguistic resources freely available to researchers
- Supports multiple corpus sizes for different computational budgets

However, limitations exist:

- **Genre diversity:** Limited to news text, missing conversational, literary, technical, and social media registers
- **No annotations:** Unannotated corpus requires separate tools for POS tagging, parsing, or semantic analysis
- **Minimal metadata:** Lack of source, temporal, and topical metadata limits certain research applications
- **Cannot study discourse:** Sentence-level segmentation without document context prevents discourse and cohesion studies
- **Temporal snapshot:** Represents only 2022 language use

Finnish Corpus Datasheet

Corpus Information

Corpus Name: Leipzig Corpora Collection - Finnish News 2022

Source: Wortschatz Leipzig [23][24]

Size: 100,000 sentences

Type: News corpus

Language Code: fin

Download: <https://wortschatz.uni-leipzig.de/en/download/fin>

Composition

Q1. What do the instances that comprise the dataset represent?

Each instance represents a single sentence extracted from Finnish-language news websites. The sentences are self-contained linguistic units automatically segmented from longer news articles using sentence boundary detection algorithms [24].

Q2. How many instances are there in total?

The corpus contains 100,000 sentences. The Leipzig Corpora Collection provides Finnish corpora in standardized sizes (10,000, 30,000, 100,000, 300,000, and 1,000,000 sentences) [23][24].

Q3. Does the dataset contain all possible instances or is it a sample?

This is a sample corpus collected from Finnish news websites via automated RSS feed monitoring during 2022 [24]. The sampling introduces several biases:

- **Genre bias:** Only news content, excluding literature, social media, blogs, and spoken language
- **Source bias:** Limited to news outlets with RSS feeds and web infrastructure, potentially excluding smaller regional Finnish-language media
- **Temporal bias:** Represents only 2022 language use, missing historical variation in the rapidly evolving Finnish language
- **Register bias:** Formal, edited news language rather than conversational Finnish

Q4. What data does each instance consist of?

Each instance consists of [23][24]:

- A unique sentence identifier
- The sentence text in Latin alphabet with Finnish diacritics (ä, ö, å)
- Pre-computed word co-occurrence statistics
- Minimal source metadata

Missing: URLs, dates, authors, article context, genre labels, topic classifications.

Q5. Is any information missing from individual instances?

Yes, substantial metadata is missing:

- **Source attribution:** No original URLs, publication dates, or news outlet identification
- **Author information:** No journalist names
- **Document context:** Sentences isolated from articles
- **Topical labels:** No classification of news type
- **Geographic origin:** No indication whether from Finnish-Finland or Swedish-speaking regions
- **Quality metadata:** No editorial quality or reliability scores

This limits analysis of source diversity, regional variation, temporal trends, and topic distribution.

Q6. Are there any errors, sources of noise, or redundancies in the dataset?

Potential sources of noise include [24]:

- **Language mixing:** Approximately 10% of web material may contain foreign language content despite filtering. Finnish news may include Swedish words (Finland is officially bilingual) and English terms.
- **Sentence segmentation errors:** Automated boundary detection may fail with:
 - Abbreviations in Finnish (e.g., "esim.", "mm.", "v.")
 - Compound words split across lines
 - Headlines and subheadings
 - Quoted speech
- **Non-sentences:** Extracted units may include:
 - Headlines
 - Photo captions
 - List items
 - Web navigation elements
- **Duplicate removal:** While exact duplicates are removed, near-duplicates (e.g., agency reports published by multiple outlets) likely remain.

- **Compound word issues:** Finnish uses extensive compounding, and segmentation may split legitimate compounds or miss word boundaries.

Q7. Does the dataset contain data that might be considered confidential?

Not applicable. All data is from publicly available news websites [23]. However, news may quote or reference individuals, and while public, this raises questions about:

- Privacy of individuals mentioned in news events
- Aggregation and permanent archival of personal information
- The "right to be forgotten" in European context

Q8. Does the dataset contain data that might be offensive or cause anxiety?

Yes, potentially. News corpora inherently contain [23]:

- **Violence and crime:** Reports of crimes, accidents, deaths
- **Political controversy:** Coverage of political conflicts and debates
- **Social issues:** Discussion of discrimination, inequality, sensitive topics
- **Negative events:** Natural disasters, economic crises, public health emergencies

The corpus creators state: "No responsibility is taken for the content of the data. The views and opinions expressed remain with the authors" [23].

Media bias: Finnish news may reflect particular political perspectives or cultural viewpoints. The corpus does not identify political leanings of sources.

Q9. Does the dataset identify any subpopulations?

Not findable. The corpus lacks systematic demographic annotations. While news discusses various groups, there is no metadata about:

- Which populations are mentioned
- Representation of Swedish-speaking vs Finnish-speaking communities
- Gender, age, or regional representation
- Indigenous Sámi population representation

Representation bias: Without demographic metadata, researchers cannot assess whether coverage is balanced across Finland's linguistic, regional, and cultural diversity.

Q10. How was the data acquired?

Data was automatically collected through web crawling [24]:

- **News collection:** Via RSS feeds from Finnish news websites on a daily basis
- **Automated crawlers:** Distributed web crawling system

- **Language filtering:** Content filtered using a language identifier based on the 5,000 most frequent Finnish words
- **Random selection:** Sentences randomly selected for standardized corpus sizes

Collection Process

Q1. Over what timeframe was the data collected?

The "Finnish News 2022" corpus was collected during 2022 through daily RSS feed monitoring [23]. However, material "may be older than the specified year" [23], potentially including late 2021 content or archived articles republished in 2022.

Temporal bias: Represents a 2022 snapshot. Major Finnish events of 2022 (e.g., NATO application, elections, COVID-19 policy) will be overrepresented. Language use may not generalize to other periods.

Q2. Were any ethical review processes conducted?

Not findable. Documentation does not mention IRB or ethical review. Implicit ethical considerations:

- Only public sources used
- No personal data collection requiring consent
- No user-generated content
- Follows principle that public web content may be collected for research

European context: May follow EU data protection principles, though this is not explicitly documented.

Q3. Does the dataset relate to people?

Yes, indirectly. News articles discuss:

- **Public figures:** Politicians, celebrities, business leaders
- **Private individuals:** People mentioned in news events
- **Communities:** References to Finnish, Swedish-speaking, Sámi, and immigrant populations

However, there is no direct data collection from individuals or first-person accounts requiring consent.

Privacy consideration: Permanent archival of names and events in a searchable corpus raises questions about privacy rights, even for publicly reported information.

Q4. Was there a specific task the dataset was created for?

Created for general-purpose linguistic analysis and NLP, not a specific task [23][24]. Intended uses:

- Word frequency analysis and lexicography
- Language modeling
- Cross-linguistic comparison
- Collocation extraction
- Finnish language resource development
- Corpus linguistics research

The standardized format enables comparison with corpora from 250+ other languages in the collection.

Preprocessing/Cleaning/Labeling

Q1. Was any preprocessing/cleaning/labeling done?

Yes, extensive preprocessing [24]:

1. **Text extraction:** Raw HTML/RSS content extracted from web pages
2. **Sentence segmentation:** Language-specific algorithms identify sentence boundaries while handling Finnish-specific punctuation patterns
3. **Language identification:** Identifier based on 5,000 most frequent Finnish words filters foreign content (note: may struggle with Finland-Swedish mixing)
4. **Non-sentence removal:** Pattern-based filtering removes fragments, navigation text, and boilerplate
5. **Duplicate removal:** Exact duplicate sentences removed
6. **Random selection:** Sentences randomly ordered and selected for standardized sizes
7. **Word tokenization:** Sentences tokenized (challenging with Finnish compounding)
8. **Co-occurrence computation:** Word co-occurrences pre-computed using log-likelihood ratio

No annotations: Corpus remains unannotated (no POS tags, morphology, or syntax).

Finnish-specific challenge: Extensive compounding and rich inflectional morphology make tokenization and word boundary detection particularly challenging.

Q2. Was the "raw" data saved in addition to preprocessed data?

No. Only processed sentences are distributed. Original web pages, HTML, or complete articles are not archived or available [23].

Limitation: Researchers cannot verify extraction accuracy, access document context, or recover article metadata.

Q3. Is the software used to preprocess/clean/label the instances available?

Partially [24][25]:

- **Methodology:** Documented in Goldhahn et al. (2012)
- **Some tools:** Available on Leipzig-Corpora-Collection GitHub (sentence cleaners, language ID)
- **TinyCC engine:** Mentioned but not fully open-source
- **Database scripts:** MySQL import scripts provided

Reproducibility: Methodology is documented but complete pipeline not fully reproducible without original crawl data.

Q36. Any other comments?

Does this corpus achieve its purpose?

Yes, for stated goals [23][24]:

- Standardized corpus enabling cross-linguistic comparison
- Clean, segmented Finnish data for NLP
- Pre-computed co-occurrence statistics
- Free resource for Finnish language research
- Multiple size options

Limitations:

- **Genre limitation:** Only news, missing conversational, literary, technical, social media Finnish
- **No morphological annotation:** Finnish's complex morphology is unannotated
- **Minimal metadata:** Lacks source, temporal, topical information
- **No document context:** Sentence-level only, preventing discourse studies
- **Temporal snapshot:** 2022 only
- **Compound handling:** Finnish compounding may be inconsistently handled

3. Corpus statistics

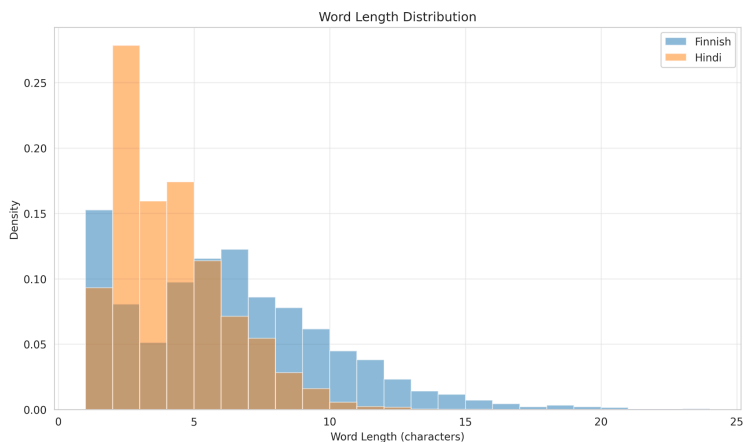
- Finnish: 394,130 unique word forms from Leipzig News corpus; 12,217 sentences (162,815 tokens) from Universal Dependencies Finnish-TDT treebank
- Hindi: 39,213 unique word forms from Leipzig News corpus; 13,306 sentences (281,057 tokens) from Universal Dependencies Hindi-HDTB treebank

Metric	Finnish	Hindi
Unique types	394,130	39,213
Total tokens	3,784,991	10,584,904
Type-token ratio	0.104	0.004
Hapax legomena	239,826 (60.85%)	0 (0.00%)
Sentences (UD)	12,217	13,306

Metric	Finnish	Hindi
Mean sentence length (words)	13.33	21.12
Std deviation (words)	9.49	9.50
10th percentile	5.00	11.00
90th percentile	23.00	34.00
Mean sentence length (chars)	86.42	80.91
Std deviation (chars)	66.35	39.10

- Mean sentence length is 13.33 words and 86.42 characters for Finnish and 21.12 words and 80.91 characters for Hindi.
 - Finnish sentences are slightly longer in characters despite fewer words. This reflects Finnish's longer average word length (6.48 vs 3.83 characters).
 - This reflects Finnish's agglutinative nature where suffixes are added to create long words. Hindi's shorter words are consistent with its moderately synthetic structure

Metric	Finnish	Hindi
Mean word length	6.48	3.83
Std deviation	4.29	2.16
10th percentile	1.00	2.00
90th percentile	12.00	7.00



- Finnish: Broader distribution, peak around 6-8 characters, long tail extending to 20+ characters. Hindi: Narrower distribution, sharp peak at 2-3 characters, tail ends around 12 characters.
- Finnish's agglutinative morphology inherently creates longer words and higher type-token ratios. Hindi's postpositional system produces shorter words with more frequent function morphemes.
- Case systems differ: Finnish uses bound morphemes (suffixes), Hindi uses free morphemes (postpositions)

Most frequent words

1. . (283,489) - period
2. , (162,767) - comma
3. on (93,207) - is/are
4. ja (85,619) - and
5. että (29,424) - that
6. ei (28,556) - not
7. oli (19,784) - was
8. mukaan (19,399) - according to
9. myös (17,136) - also
10. ovat (16,079) - are

1. के (417,400) - of/possessive marker
2. . (375,142) - period
3. में (324,300) - in
4. की (253,645) - of/possessive marker (feminine)
5. है (240,716) - is
6. को (201,336) - to/dative marker
7. से (175,324) - from/instrumental marker
8. , (172,730) - comma
9. ने (172,108) - ergative marker
10. और (157,998) - and

- Hindi shows higher frequency of postpositions/case markers (के, में, की, को, से, ने)
- Finnish copula forms ("on", "oli", "ovat") are highly frequent

Bigram analysis

1. , että (1,076) - ", that"	1. है I (4,283) - "is [period]"
2. , mutta (435) - ", but"	2. कहा कि (1,713) - "said that"
3. , ja (434) - ", and"	3. के लिए (1,639) - "for"
4. , joka (335) - ", which"	4. हैं I (1,450) - "are [period]"
5. ei ole (224) - "is not"	5. है कि (1,272) - "is that"

- Finnish bigrams show comma-initial patterns reflecting subordinate clause structures
- Hindi bigrams frequently involve postposition combinations (के लिए = for)

Trigram analysis

1. , että ei (74) - ", that not"	1. ने कहा कि (779) - "X said that"
2. siitä , että (61) - "from that, that"	2. उन्होंने कहा कि (597) - "he/she said that"
3. , joka on (55) - ", which is"	3. गया है I (377) - "has gone [period]"
4. (EY) N:o (55) - "(EU) No" [legal documents]	4. रहा है I (273) - "is happening [period]"

- Both reflect news corpus conventions

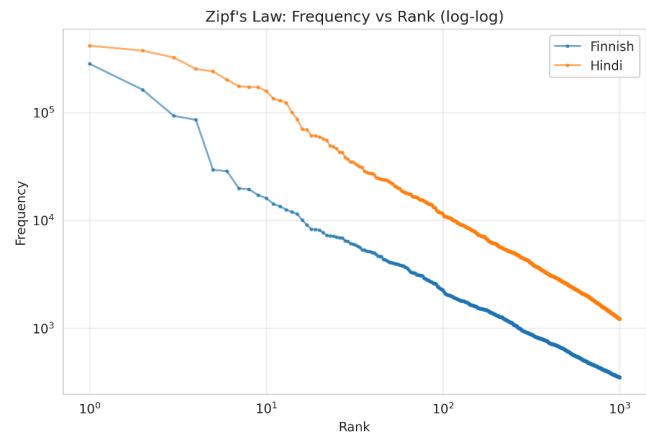
Conjunctions

Language	Conjunction	Frequency	Percentage
Finnish	ja (and)	85,619	2.26%
	tai (or)	6,462	0.17%
	mutta (but)	14,170	0.37%
	että (that)	29,424	0.78%
	kun (when)	11,440	0.30%
Hindi	और (and)	157,998	1.49%
	या (or)	10,762	0.10%
	लेकिन (but)	27,861	0.26%
	परन्तु (but-formal)	54	0.0005%
	किन्तु (but-literary)	18	0.0002%

- Hindi has multiple formal/literary variants for "but" (लेकिन, परन्तु, किन्तु), but only colloquial लेकिन is common
- The percentages are way higher for Finnish, because there are fewer distinct words due to its synthetic nature.

Zipf's Law

Both languages follow Zipf's law on log-log plots, with frequency inversely proportional to rank. The curves are nearly parallel, indicating similar power-law distributions despite different morphological structures.



Deliverable 4: Morphological Analysis with Sub-word Tokenizers

Methodology

We trained Byte Pair Encoding (BPE) tokenizers for both Finnish and Hindi using the Hugging Face tokenizers library with a vocabulary size of 5,000 tokens.

Training Data:

- Finnish: 162,815 words from 12,217 sentences (UD Finnish-TDT)
- Hindi: 281,057 words from 13,306 sentences (UD Hindi-HDTB)

Test Data:

- Finnish: 21,070 words from 1,555 sentences
- Hindi: 35,430 words from 1,684 sentences

Design Decisions

NFKC Normalization: Applied ✓

Converts characters to standard Unicode form to ensure consistency across different encodings.

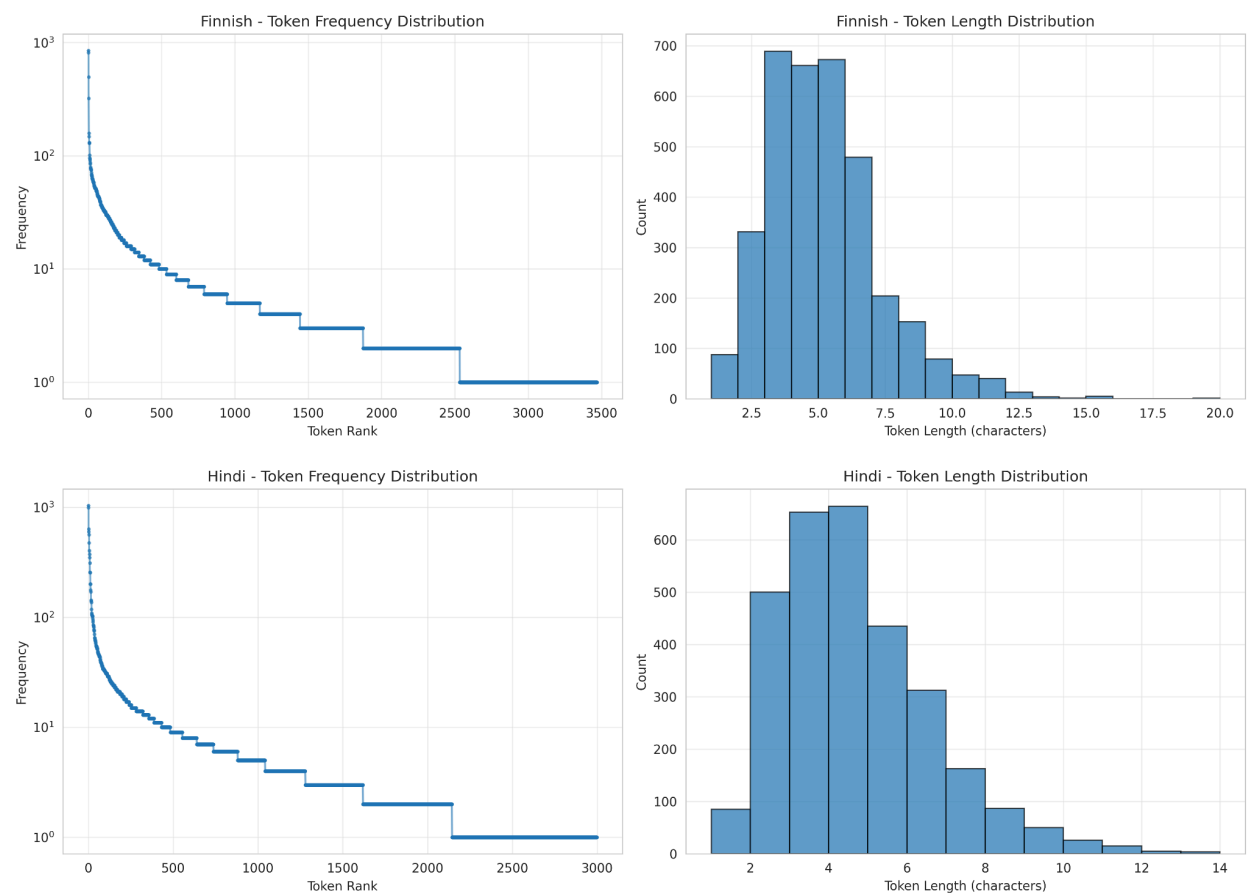
Strip Accents: NOT Applied ✗

Finnish requires ä and ö distinctions (different vowels with different meanings). Hindi Devanagari diacritics are essential for meaning. Removing accents would destroy linguistic information.

Lowercase: NOT Applied ✗

Preserves proper nouns and sentence structure, which are important in news corpora.

Tokenization Statistics



Finnish Results:

Word Category	Tokens/Word	Avg Token Length
Frequent (≥ 100)	1.00	4.07 chars
Rare (2-5)	2.38	3.64 chars
Hapax (1)	3.34	3.35 chars
Unseen	3.42	3.36 chars

Finnish test corpus: 23,897 total tokens, 3,469 unique tokens, average 1.77 tokens per word.

Top 10 Finnish tokens:

1. . (849)
2. , (815)
3. ja (495) - and
4. on (322) - is
5. että (158) - that

6. se (148) - it
7. - (131)
8. a (130)
9. ei (129) - not
10. t (101)

Hindi Results:

Word Category	Tokens/Word	Avg Token Length
Frequent (≥ 100)	1.00	3.38 chars
Rare (2-5)	2.42	2.34 chars
Hapax (1)	2.83	2.26 chars
Unseen	3.06	2.21 chars

Hindi test corpus: 25,629 total tokens, 3,000 unique tokens, average 1.20 tokens per word.

Top 10 Hindi tokens:

1. के (1,037) - possessive
2. । (992) - devanagari period
3. में (635) - in
4. की (602) - possessive (fem)
5. है (565) - is
6. को (476) - dative marker
7. ने (404) - ergative marker
8. कि (375) - that
9. से (349) - from/instrumental
10. का (312) - possessive

Key Finding: Hindi achieves better compression (1.20 tokens/word) than Finnish (1.77 tokens/word) due to shorter words and simpler morphology.

Morphological Segmentation Examples

Finnish - Plurals (-t marker):

- talo → talo (house)
- talot → talo | t ✓ (houses - correctly segmented)
- kissat → kissa | t ✓ (cats - correctly segmented)

Finnish - Case markers:

- talossa → talo | ssa ✓ (in the house)
- talosta → talo | sta ✓ (from the house)

- taloon → talo | on ✓ (into the house)
- talon → talon ✗ (genitive not segmented)

Hindi - Postpositions (separate words):

- घर में → घर | में ✓ (in house)
- घर से → घर | से ✓ (from house)
- घर को → घर | को ✓ (to house)

Segmentation Consistency Analysis

Finnish - Words ending in '-t' (potential plurals):

Consistency varies:

- meidät → mei | dät ✗ (pronoun, not plural)
- ihmiset → ihmiset ✗ (people, plural not segmented)
- uskaltautuivat → uskalta | utuivat ✓ (verb past plural)
- onnistunut → onnistunut ✗ (participle -nut, not plural)
- tapahtumat → tapahtu | mat ✓ (events, plural segmented)
- alkoivat → alkoi | vat ✓ (began, plural verb)

Interpretation:

- High-frequency words (ihmiset) learned as single tokens
- Verb plural markers (-vat/-ivat) more consistently segmented
- Participles (-nut) not segmented as plural despite -t ending

Hindi: Hindi postpositions are nearly always segmented correctly as separate tokens because they are:

1. Free morphemes (separate words)
2. High-frequency items in the corpus
3. Consistently space-separated in Devanagari text

Analysis

Why BPE doesn't follow morphological intuition:

1. **Statistical vs. linguistic:** BPE optimizes for compression, not morphological correctness. High-frequency words learned as single units even if inflected.
2. **Frequency effects:** Common words like "ihmiset" (people) stay whole despite having plural marker.

3. **Cannot generalize patterns:** Finnish plural can be -t, -i, -it depending on the word. BPE sees these as different patterns.

Why Hindi works better:

1. **Separate words for grammar:** Postpositions are naturally separated (घर में), making segmentation easier.
2. **Shorter words:** Less internal complexity to segment (3.83 chars avg vs Finnish 6.48).
3. **High-frequency markers:** Postpositions like के, में, को appear thousands of times as distinct tokens.

Impact on Applications

For Finnish: Machine translation may struggle with new inflected forms. Search engines may miss related word forms without additional stemming.

For Hindi: Better performance due to postpositional structure being preserved. More natural text generation.

Deliverable 5: Dependency Parsing and Syntactic Analysis

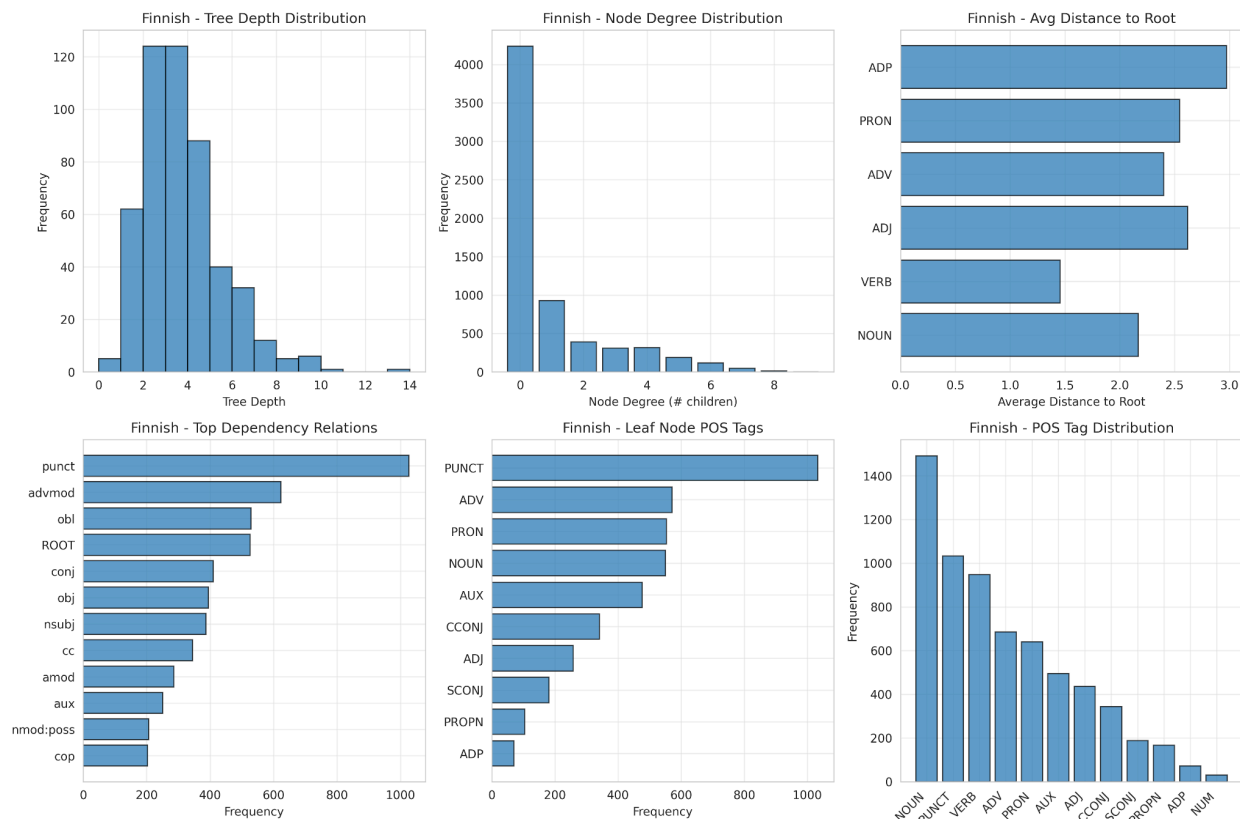
Methodology

Parser: SpaCy fi_core_news_sm (Finnish only - no pre-trained Hindi parser available)

Training data: Universal Dependencies Finnish-TDT

Evaluation data: 500 sentences from UD Finnish-TDT test split

Dependency Tree Statistics



Metric	Value
Average tree depth	3.25
Standard deviation	1.79
Min depth	0
Max depth	13

Finnish dependency trees are relatively shallow with most sentences having 3-4 levels of syntactic embedding.

Node Degree Distribution:

- Degree 0 (leaves): 4,236 nodes (64.66%)
- Degree 1: 926 nodes (14.14%)
- Degree 2: 391 nodes (5.97%)

Nearly two-thirds of nodes are leaves with no children.

Average Distance to Root by POS:

POS Tag	Distance
VERB	1.46
AUX	1.91
PUNCT	1.98
NOUN	2.17
ADV	2.40
PRON	2.54

Verbs are closest to root (1.46), consistent with verb-headed structures. Nouns are more distant (2.17) as they're typically dependents of verbs.

Most Common Leaf Nodes:

POS Tag	Count	Percentage
PUNCT	1,033	24.39%
ADV	571	13.48%
PRON	553	13.05%
NOUN	551	13.01%
AUX	476	11.24%

Punctuation is the most common leaf node (24.39%) as it never has syntactic dependents. High frequency of adverbs and pronouns as leaves reflects Finnish sentence structure.

Most Common Dependency Relations:

Relation	Count	Percentage
punct	1,027	15.68%
advmod	623	9.51%
obl	529	8.08%
ROOT	526	8.03%
conj	410	6.26%
obj	395	6.03%
nsubj	387	5.91%

High frequency of 'advmod' (9.51%) shows Finnish uses many adverbials. 'obl' (oblique, 8.08%) reflects the rich case system with many locative and instrumental constructions. 'conj' (6.26%) indicates frequent coordination.

Example Dependency Parses

Example 1: Simple sentence

"Taas teatteriin" (Again to-theater)

Token	POS	Head	Dep Rel	Children
Taas	ADV	teatteriin	advmod	
teatteriin	VERB	ROOT	ROOT	Taas

Analysis: Very short sentence. "teatteriin" (to theater) functions as ROOT, with "taas" (again) as adverbial modifier.

Example 2: Moderate complexity

"Tänäänkin pitäisi mennä teatteriin ." (Today-too should go to-theater .)

Token	POS	Head	Dep Rel	Children
Tänäänkin	ADV	mennä	advmod	
pitäisi	AUX	mennä	aux	
mennä	VERB	ROOT	ROOT	Tänäänkin, pitäisi, teatteriin, .
teatteriin	NOUN	mennä	obl	
.	PUNCT	mennä	punct	

Analysis:

- Main verb "mennä" (to go) is ROOT
- Modal auxiliary "pitäisi" (should) attached as **aux**
- Adverb "tänäänkin" (today too) attached as **advmod**
- Destination "teatteriin" (to theater) marked as **obl** (oblique)

Ambiguous Sentence Testing

Test 1: *"Näen miehen kaukoputkella."* (I-see man-ACC telescope-ADE)

Ambiguous: (1) I see the man with the telescope (man has telescope), or (2) I see the man using the telescope (I use telescope)

Token	POS	Head	Dep Rel
Näen	VERB	ROOT	ROOT
miehen	NOUN	kaukoputkella	nmod:poss
kaukoputkella	NOUN	Näen	obl

Parser choice: "kaukoputkella" (telescope) attaches to verb "Näen" as **obl** (oblique), with "miehen" possessor of telescope.

Interpretation: Parser chose reading (1): man with telescope. Adessive case (-lla) on "kaukoputkella" suggests possession/association.

Test 2: "Poliisi ampui miehen aseella." (Police shot man-GEN gun-ADE)

Ambiguous: (1) Police shot the man with the man's gun, or (2) Police shot the man using a gun

Token	POS	Head	Dep	Rel
Poliisi	NOUN	ampui	nsubj	
ampui	VERB	ROOT	ROOT	
miehen	NOUN	aseella	nmod:poss	
aseella	NOUN	ampui	obl	

Parser choice: "aseella" (gun) attaches to verb "ampui" as oblique, with "miehen" as possessor.

Interpretation: Parser chose reading (1): man's gun. Again, genitive "miehen" and adessive "aseella" suggest possessive relation.

Key Findings

- Finnish dependency trees are relatively shallow (avg depth 3.25)
- Verbs are central, closest to root
- High frequency of oblique relations reflects rich case system
- Parser uses case marking consistently for attachment decisions
- Flexible word order creates genuine ambiguities in parsing

Part C: Reflection

For Hindi, we were not sure about the cause of the 0% hapax legomena rate. The absence of a SpaCy Hindi dependency parser further restricted syntactic comparisons. Devanagari encoding also required careful handling to prevent text corruption. At the corpus level, relying solely on news data made the lack of broader linguistic variation. Our statistical analysis revealed the linguistic differences we had noticed during our background research—showing the difference between Hindi’s analytic structure and Finnish’s agglutinative morphology through lexical diversity and word length.

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